UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Introduction to Higher-Order Linear Differential Equations

Complete study pack for class 001: linear differential equations of order n, constant and variable coefficients, homogeneous and non-homogeneous equations, the operator D and $f(D)$, linear independence, the complementary function and the particular integral

Compiled from the PESU Unit 2 class 001 slide deck and AV-summary transcript, and the Unit 2 questions of eight past ESA papers (via the shared ESA map). Generated 15 October 2026.

· How to use this pack

Topic 1 defines a linear DE and its order, topic 2 classifies linear DEs (constant or variable coefficients; homogeneous or not), topic 3 introduces the operator D and $f(D)$, topic 4 explains linear independence and why the complementary function is $c_1y_1 + \cdots + c_ny_n$, topic 5 puts it together: $y = \mathrm{CF} + \mathrm{PI}$. This class has no methods of solution yet. How to find the CF from the roots of the auxiliary equation (all cases) is in the class 002 pack "Solutions of Homogeneous Linear DEs: Complementary Functions"; the PI rules are in classes 003–006. Figure F5.1 is the plan of every Unit 2 answer. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·12	Unit 2 deck 001, slide 12 (rendered .pptx)
T001 00:17:38	Unit 2 AV-summary transcript 001 ("Introduction to higher order differential equations, complementary function and particular integral"), segment start
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s3 3d	Past paper: UE23MA141A Jan 2024 set 3, question 3.d

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1 Linear differential equations and their order

An ordinary differential equation is either linear or non-linear. It is *linear* when the dependent variable y and its derivatives occur only in the first degree and are not multiplied together; the coefficients may depend on x but not on y. The *order* is the order of the highest derivative present. The deck writes the general linear equation of order 1, 2 and n; Unit 2 is about orders 2 and higher.

Key terms and definitions

Linear differential equation (slide). A differential equation in which the dependent variable and its derivative occur in first degree and are not multiplied together is a Linear Differential Equation. S001·6 · T001 00:05:13

Linear differential equation (lecture form). A differential equation is linear if (1) the coefficients of y and its derivatives are functions of x or constants (never of y), and (2) the degree of y and of each derivative is 1. T001 00:06:27 · T001 00:06:41

Non-linear differential equation. A differential equation which is not linear, e.g. $y^2\frac{dy}{dx} + \sin y = x^2 e^{2y}$. S001·5 · T001 00:03:29

Order. The order of the highest-order derivative present in the differential equation. T001 00:05:51

Higher-order LDE. A linear differential equation of order 2 or more: $\frac{d^n y}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \cdots + a_n y = X$, where $a_1, a_2, \ldots, a_n$ are functions of x or constants and X is a function of x alone. S001·8 · T001 00:09:46

> *"A linear differential equation of first order is of the form $\frac{dy}{dx} + P\cdot y = Q$ where P and Q are functions of x alone."* S001·7

> *"A linear differential equation of order two is of the form $\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2 y = X$ where P_1 and P_2 and X are functions of x alone."* S001·7

> *"A linear differential equation of order n is of the form $\frac{d^n y}{dx^n} + P_1\frac{d^{n-1}y}{dx^{n-1}} + P_2\frac{d^{n-2}y}{dx^{n-2}} + \cdots + P_n y = X$ where $P_1, P_2, \ldots, P_n$ and X are functions of x alone."* S001·7

Formulas

E1.1 · First-order linear DE ESA ★

$$\frac{dy}{dx} + P\,y = Q$$

gives	The general linear equation of order 1.
applies	P, Q functions of x alone.
note	Solved in the earlier unit by an integrating factor; not part of Unit 2.

S001·6 · S001·7 · T001 00:07:04

E1.2 · Second-order linear DE

ESA ★★

$$\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2\,y = X$$

GIVES The general linear equation of order 2.

APPLIES P_1, P_2, X functions of x alone.

NOTE The lecture writes $P_0y'' + P_1y' + P_2y = X$ with $P_0 \neq 0$; dividing by P_0 gives the slide's form.

S001·7 · T001 00:07:22

E1.3 · Linear DE of order n

ESA ★★

$$\frac{d^ny}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \cdots + a_ny = X$$

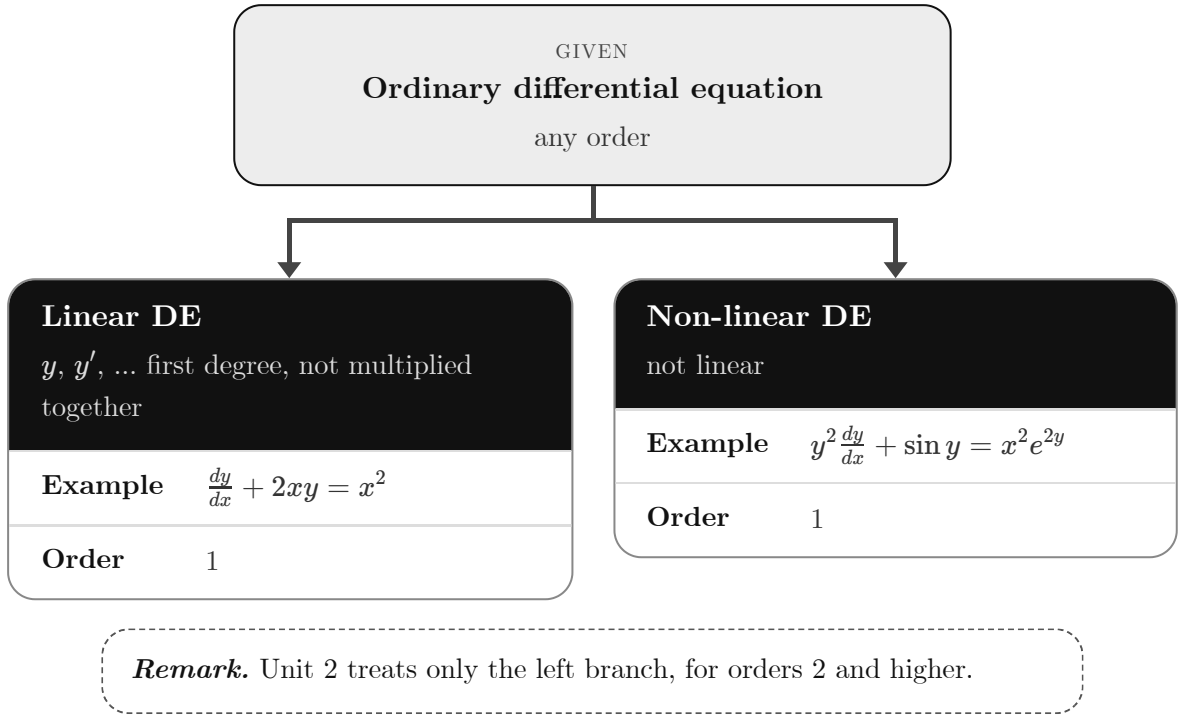
GIVES The general linear equation of order n.

APPLIES $a_1, \ldots, a_n$ functions of x or constants; X a function of x alone.

NOTE Slide 001·7 writes $P_1, \ldots, P_n$; slide 001·8 onward $a_1, \ldots, a_n$.

S001·7 · S001·8 · T001 00:08:38 · T001 00:09:46

Flowcharts



Remark. Unit 2 treats only the left branch, for orders 2 and higher.

Figure F1.1. Classifying an ordinary differential equation (slide 001·5) ESA ★

S001·5 · T001 00:02:54 · T001 00:03:46

Important points and why

1. Linear means: y and its derivatives only to the first degree, never multiplied together, and coefficients free of y. $y\frac{dy}{dx}$, y^2, $\sin y$, e^y make an equation non-linear. ESA ★ S001·5 · S001·6 · T001 00:05:18 · T001 00:05:34
2. The order is the highest derivative present, not the highest power. ESA ★ T001 00:05:51 · T001 00:11:19
3. Unit 2 treats only linear equations of order 2 and higher; first-order and non-linear equations were the previous unit. ESA ★ T001 00:04:04 · T001 00:04:22

Exam trap. The lecture says "P_0 equal to 0" for the second-order form (T001 00:07:34); it must be $P_0 \neq 0$, otherwise the equation is not of order 2. T001 00:07:34 · T001 00:08:55

Worked examples

W1.1 · Linear or non-linear? (slide 001·5) ESA ★

GIVEN Classify (a) $\frac{dy}{dx} + 2xy = x^2$ and (b) $y^2\frac{dy}{dx} + \sin y = x^2e^{2y}$.

STEP 1 (a) y and $\frac{dy}{dx}$ occur to the first degree, not multiplied together; the coefficient $2x$ is a function of x only.

STEP 2 (b) $y^2\frac{dy}{dx}$ multiplies y (squared) by its derivative; $\sin y$ and e^{2y} are not first degree in y.

ANSWER (a) linear, order 1; (b) non-linear, order 1.

CHECK Both definitions (slide 001·6 and the lecture's two conditions) agree.

S001·5 · T001 00:03:46 · T001 00:05:18

SHORT-NOTE RECAP

A DE is linear when y and its derivatives occur only in the first degree and are not multiplied together; coefficients may be functions of x or constants, never of y. Order = highest derivative. Linear of order n: $y^{(n)} + a_1y^{(n-1)} + \cdots + a_ny = X$. Example: $\frac{dy}{dx} + 2xy = x^2$ is linear; $y^2\frac{dy}{dx} + \sin y = x^2e^{2y}$ is not.

2 Constant or variable coefficients; homogeneous or not

A linear DE of order n, $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_n y = X$, comes in two forms: with constant coefficients, where all a_i are numbers, or with variable coefficients, where some a_i are functions of x. An equation with constant coefficients is *homogeneous* when the right-hand side X is zero and *non-homogeneous* when X is a nonzero function of x. Every non-homogeneous equation has an associated homogeneous equation, obtained by putting $X = 0$.

Key terms and definitions

LDE with constant coefficients. In this type, the coefficient of derivatives are constants. S001·8 · T001 00:10:28

LDE with variable coefficients. In this type, the coefficient of derivatives $a_1, a_2, \ldots, a_n$ are functions of x. S001·9 · T001 00:11:33

Homogeneous LDE. RHS X is zero. S001·10 · T001 00:13:22

Non-homogeneous LDE. RHS X is a nonzero function of x. S001·10 · T001 00:13:34

Associated homogeneous equation. The equation obtained from a non-homogeneous LDE by replacing X by 0: for $f(D)y = X$ it is $f(D)y = 0$. T001 00:14:44 · T001 00:14:52

Formulas

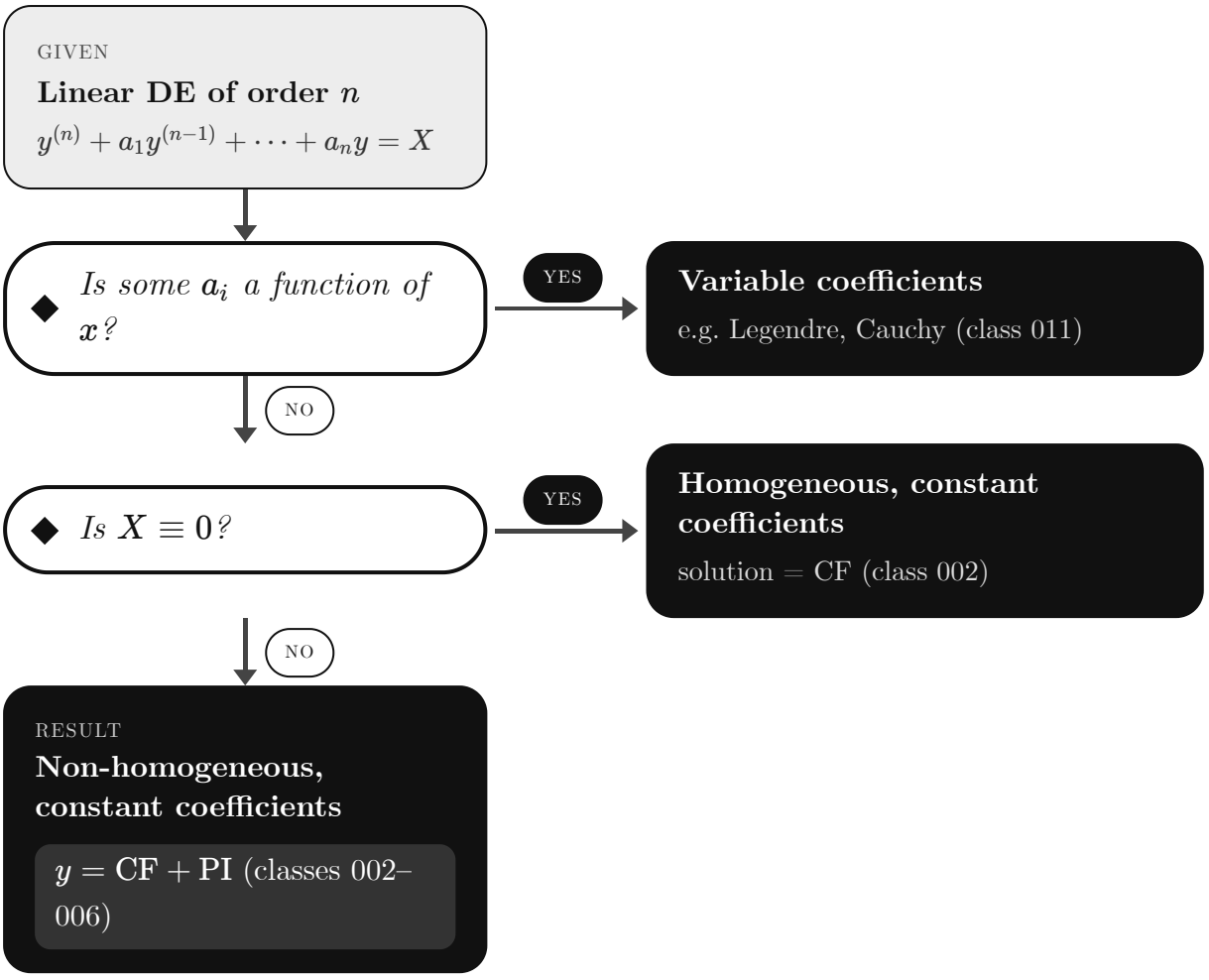
E2.1 · Homogeneous and non-homogeneous ESA ★★

$$X = 0 : \text{ homogeneous}; \qquad X \not\equiv 0 : \text{ non-homogeneous}$$

gives	The type of the equation from its right-hand side.
applies	Any LDE (the deck states it for constant coefficients).
note	Slide 001·10: $2y'' - 3y' + 9y = 0$ (homogeneous) and $2y'' - 3y' + 9y = x \sinh x$ (non-homogeneous).

S001·10 · T001 00:13:22 · T001 00:13:52

Flowcharts



Remark. Slide 001·10 draws the homogeneous / non-homogeneous split as a tree under "LDE with constant coefficients"; the example pair is $2y'' - 3y' + 9y = 0$ and $2y'' - 3y' + 9y = x \sinh x$.

Figure F2.1. Classifying a linear DE (slides 001·8–001·10) ESA ★★

S001·8 · S001·9 · S001·10 · T001 00:10:11 · T001 00:13:13

The forms of a linear DE in the deck

Form	Coefficients	RHS X	Deck example	Order
First-order linear	P function of x	$Q(x)$	$\frac{dy}{dx} + 2xy = x^2$	1
Constant coefficients	numbers	x^2	$2y''' - 5y'' + 7y' + y = x^2$	3
Variable coefficients	functions of x	$x^2 \sin x$	$2x^3y''' - (5x^2 + 2)y'' + xy' + y = x^2 \sin x$	3
Homogeneous (const.)	numbers	0	$2y'' - 3y' + 9y = 0$	2
Non-homogeneous (const.)	numbers	$x \sinh x$	$2y'' - 3y' + 9y = x \sinh x$	2

S001·7 · S001·8 · S001·9 · S001·10

Important points and why

1. Classify before solving: constant coefficients (classes 002–006, operator methods) or variable coefficients (class 011: Legendre and Cauchy forms, 8 ESA occurrences). Homogeneous (CF only) or non-homogeneous (CF + PI). ESA ★★ S001·3 · S001·8 · S001·9 · ESA Jan-24 s3 3d
2. A constant term or a function of x on the right makes the equation non-homogeneous; only $X \equiv 0$ is homogeneous. ESA ★★ S001·10 · T001 00:13:34
3. Every non-homogeneous LDE has an associated homogeneous equation: the same left side with X replaced by 0. ESA ★★ T001 00:14:27 · T001 00:14:52

Exam trap. $2x^3y''' - (5x^2 + 2)y'' + xy' + y = x^2 \sin x$ is linear (variable coefficients). Functions of x in the coefficients do not make an equation non-linear; only y in a coefficient does. S001·9 · T001 00:12:08

Worked examples

W2.1 · Classify the deck's four equations ESA ★★

given Give order, type of coefficients and homogeneity: (a) $2y''' - 5y'' + 7y' + y = x^2$; (b) $2x^3y''' - (5x^2+2)y'' + xy' + y = x^2\sin x$; (c) $2y'' - 3y' + 9y = 0$; (d) $2y'' - 3y' + 9y = x\sinh x$.

step 1 (a) Coefficients $2, -5, 7, 1$ are numbers; highest derivative y'''; $X = x^2 \neq 0$.

step 2 (b) Coefficients $2x^3,\ -(5x^2+2),\ x,\ 1$ are functions of x; order 3; $X = x^2\sin x$.

step 3 (c) Constants $2, -3, 9$; order 2; $X = 0$. (d) Same left side; $X = x\sinh x \neq 0$, and (c) is its associated homogeneous equation.

answer (a) order 3, constant coefficients, non-homogeneous; (b) order 3, variable coefficients, non-homogeneous; (c) order 2, constant, homogeneous; (d) order 2, constant, non-homogeneous.

check sympy: the general solution of (a) has 3 constants, matching order 3; $y = e^{3x/4}\cos\frac{3\sqrt{7}x}{4}$ satisfies (c) (roots of $2m^2 - 3m + 9 = 0$ are $\frac{3}{4} \pm \frac{3\sqrt{7}}{4}i$).

S001·8 · S001·9 · S001·10 · T001 00:11:01 · T001 00:12:26 · T001 00:14:09

short-note recap

Two forms: constant coefficients ($2y''' - 5y'' + 7y' + y = x^2$) and variable coefficients ($2x^3y''' - (5x^2+2)y'' + xy' + y = x^2\sin x$). Homogeneous: $X = 0$ ($2y'' - 3y' + 9y = 0$); non-homogeneous: X a nonzero function of x ($2y'' - 3y' + 9y = x\sinh x$). Putting $X = 0$ gives the associated homogeneous equation.

3 The differential operator D and $f(D)$

Writing $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, ..., $D^n = \frac{d^n}{dx^n}$ turns the equation into $f(D)y = X$, where $f(D) = D^n + a_1 D^{n-1} + \cdots + a_n$ is the *linear differential operator*. With constant coefficients $f(D)$ is a polynomial in D: it can be factorised and its factors taken in any order. Every constant-coefficient ESA question of Unit 2 is written this way.

Key terms and definitions

Differential operator D. $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, $D^3 = \frac{d^3}{dx^3}$, ..., $D^n = \frac{d^n}{dx^n}$. S001·11 · T001 00:15:16

Linear differential operator $f(D)$. $f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n$, so that the LDE reads $f(D)y = X$. S001·11 · T001 00:16:29

> *"Thus, in terms of operator notation $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = X$ can be written as $D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + \cdots + a_n y = X$, $f(D)y = X$, where $f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n$ called linear differential operator."* S001·11

Formulas

E3.1 · Operator notation ESA ★★★

$$D = \frac{d}{dx}, \quad D^2 = \frac{d^2}{dx^2}, \quad D^3 = \frac{d^3}{dx^3}, \ \ldots, \ D^n = \frac{d^n}{dx^n}$$

GIVES Shorthand for derivatives: $Dy = y'$, $D^2 y = y''$.

APPLIES Always; ESA questions use it.

NOTE $D^2 y$ means $\frac{d^2 y}{dx^2}$, not $\left(\frac{dy}{dx}\right)^2$.

S001·11 · T001 00:15:16

E3.2 · The equation as $f(D)y = X$ ESA ★★★

$$f(D)\, y = X, \qquad f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n$$

GIVES The compact form used in every later class.

APPLIES Constant a_i: $f(D)$ is a polynomial in D.

NOTE a_n multiplies y itself ($D^0 = 1$).

S001·11 · S001·12 · T001 00:16:29

E3.3 · Factorising $f(D)$ (constant coefficients) ESA ★★

$$(D-\alpha)(D-\beta)y = (D-\beta)(D-\alpha)y = \big[D^2-(\alpha+\beta)D+\alpha\beta\big]y$$

GIVES $f(D)$ may be factorised into linear factors, taken in any order.

APPLIES α, β constants.

NOTE "It obeys the laws of algebra" (T001 00:16:49). Fails for variable coefficients.

T001 00:16:49 · T001 00:17:07

E3.4 · Preview: $f(D)$ acting on e^{mx} ESA ★★★

$$D^k e^{mx} = m^k e^{mx} \implies f(D)e^{mx} = f(m)e^{mx}$$

GIVES e^{mx} solves $f(D)y=0$ exactly when $f(m)=0$, the auxiliary equation.

APPLIES Constant coefficients.

NOTE Not in this deck; the auxiliary equation and all CF cases are class 002.

ADDED FOR CLARITY S001·16

Important points and why

1. Constant-coefficient ESA questions are written as $f(D)y = X$, e.g. $(D-2)^2y = 8(e^{2x}+\sin 2x)$ and $(D^3-6D^2+11D-6)y = e^{2x}+x$; read D as $\frac{d}{dx}$. ESA ★★★ S001·11 · ESA Dec-23 3a · ESA Jan-24 s2 2d
2. $f(D)$ with constant coefficients behaves like an algebraic polynomial: factorise it into linear factors, in any order. ESA ★★ T001 00:16:49 · T001 00:17:07

Why is the operator notation useful?
With constant coefficients $f(D)$ is a polynomial: it can be factorised, and (class 002) $f(D)e^{mx} = f(m)e^{mx}$ turns the DE into an algebraic equation. Classes 003–006 also invert it, $y_p = \frac{1}{f(D)}X$.

ADDED FOR CLARITY S001·11 · T001 00:16:49

Exam trap. $f(D)$ commutes and factorises like a polynomial only with constant coefficients: $D(xDy) = xD^2y + Dy \neq xD(Dy)$. ADDED FOR CLARITY T001 00:16:49

Exam trap. D^2y is $\frac{d^2y}{dx^2}$, not $\left(\frac{dy}{dx}\right)^2$; and the last coefficient a_n multiplies y, so $(D^2+4)y = y''+4y$. ADDED FOR CLARITY S001·11

Worked examples

W3.1 · Write in operator form ESA ★★★

GIVEN Write (a) $2\frac{d^3y}{dx^3} - 5\frac{d^2y}{dx^2} + 7\frac{dy}{dx} + y = x^2$ and (b) $y''' - 6y'' + 11y' - 6y = e^{2x} + x$ as $f(D)y = X$.

STEP 1 Replace $\frac{d^ky}{dx^k}$ by D^ky and take y out: (a) $(2D^3 - 5D^2 + 7D + 1)y = x^2$.

STEP 2 In the slide's normalised form divide by 2: $\left(D^3 - \frac{5}{2}D^2 + \frac{7}{2}D + \frac{1}{2}\right)y = \frac{1}{2}x^2$.

STEP 3 (b) $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$, exactly as ESA Jan-24 s2 2d is printed; $f(D) = (D-1)(D-2)(D-3)$.

ANSWER (a) $f(D) = 2D^3 - 5D^2 + 7D + 1$, $X = x^2$. (b) $f(D) = D^3 - 6D^2 + 11D - 6 = (D-1)(D-2)(D-3)$, $X = e^{2x} + x$.

CHECK Expanding $(D-1)(D-2)(D-3)$ gives $D^3 - 6D^2 + 11D - 6$ (sympy).

S001·8 · S001·11 · ESA Jan-24 s2 2d

W3.2 · Factors commute (constant coefficients only) ESA ★★

GIVEN Show $(D-1)(D-2)u = (D-2)(D-1)u = (D^2 - 3D + 2)u$ for any $u(x)$, and that $D(xD) \neq (xD)D$.

STEP 1 $(D-2)u = u' - 2u$; then $(D-1)(u' - 2u) = u'' - 2u' - u' + 2u = u'' - 3u' + 2u$.

STEP 2 $(D-1)u = u' - u$; then $(D-2)(u' - u) = u'' - u' - 2u' + 2u = u'' - 3u' + 2u$. Same.

STEP 3 Variable coefficient: $D(xu') = u' + xu''$, but $xD(u') = xu''$. They differ by u'.

ANSWER Constant-coefficient factors commute; $D(xD) - (xD)D = D$.

CHECK sympy: difference 0 in the first case and u' in the second.

ADDED FOR CLARITY T001 00:16:49 · T001 00:17:07

Past-paper questions

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q2.d · solved in W3.1

Solve $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$. (Here: reading $f(D)$; full solution in class 005.)

SHORT-NOTE RECAP

$D = \frac{d}{dx}$, $D^n = \frac{d^n}{dx^n}$. The LDE becomes $f(D)y = X$ with $f(D) = D^n + a_1D^{n-1} + \cdots + a_n$, the linear differential operator. With constant coefficients $f(D)$ is a polynomial in D: factorise it, in any order. Preview: $f(D)e^{mx} = f(m)e^{mx}$ (class 002).

4 Linear independence and the complementary function

Functions $f_1, \ldots, f_n$ are linearly dependent on an interval when some combination $c_1 f_1 + \cdots + c_n f_n$ with constants not all zero vanishes identically; otherwise they are independent. Two functions are independent when neither is a constant multiple of the other. If $y_1, \ldots, y_n$ are n independent solutions of an n-th order homogeneous LDE, then $c_1 y_1 + \cdots + c_n y_n$ is its complete solution, the *complementary function* (CF).

Key terms and definitions

Linearly dependent functions. A set of functions $f_1(x), f_2(x), \ldots, f_n(x)$ are said to be linearly dependent on an interval I if there exists constants $c_1, c_2, \ldots, c_n$ not all zero, such that $c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0$. S001·13 · T001 00:19:05

Linearly independent functions. If not linearly dependent, the functions are called linearly independent: $c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0 \Rightarrow c_1 = c_2 = \cdots = c_n = 0$. S001·13 · T001 00:19:59

Complementary function (CF). The complete solution $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ of the homogeneous LDE of order n, where $y_1, \ldots, y_n$ are n linearly independent solutions. S001·12 · S001·16 · T001 00:25:38

> *"Two functions y_1 and y_2 are linearly independent when $c_1 y_1 + c_2 y_2 \neq 0$ for non zero constants c_1 and c_2, i.e. $c_1 y_1 \neq -c_2 y_2 \;\rightarrow\; y_1 \neq \left(\frac{-c_2}{c_1}\right) y_2$."* S001·14

> *"If y_1 and y_2 are two linearly independent solutions of a second order homogeneous linear differential equation, then $c_1 y_1 + c_2 y_2$ is also a solution of the differential equation. It is the complete solution of the differential equation."* S001·15

> *"If the nth order LDE with constant coefficients $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = 0$ has n linearly independent solutions $y_1, y_2, y_3, \ldots, y_n$, then their linear combination $c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ is also a solution of the differential equation. $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$, the complete solution is called COMPLEMENTARY FUNCTION of the differential equation."* S001·16

Formulas

E4.1 · Linear dependence and independence ESA ★

$$\text{independent:}\quad c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0 \;\Rightarrow\; c_1 = c_2 = \cdots = c_n = 0$$

gives	The test for a set of functions.
applies	On an interval I; "$= 0$" means for every x in I.
note	Dependent: some constants not all zero make the combination vanish.

S001·13 · T001 00:19:05 · T001 00:20:17

E4.2 · Two functions

ESA ★★

$$y_1,\ y_2 \text{ independent} \iff \frac{y_1}{y_2} \neq \text{constant}$$

GIVES A quick test for a pair.

APPLIES Two functions only.

NOTE Neither is a constant multiple of the other (T001 00:22:00).

S001·14 · T001 00:21:01 · T001 00:22:00

E4.3 · Wronskian of two functions

ESA ★

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

GIVES $W \neq 0$ (at some point of I) shows y_1, y_2 independent; for two solutions of the same LDE, $W \equiv 0$ means dependent.

APPLIES For n functions: the $n \times n$ determinant of the functions and their first $n-1$ derivatives.

NOTE Not in the deck. $W(e^x, e^{2x}) = e^{3x}$, $W(e^x, 3e^x) = 0$. Used in class 011 (variation of parameters).

ADDED FOR CLARITY S001·14

E4.4 · Superposition: the complementary function

ESA ★★★

$$y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

GIVES The complete solution of the homogeneous LDE $f(D)y = 0$ of order n.

APPLIES $y_1, \ldots, y_n$ are n linearly independent solutions.

NOTE Second order: $y_c = c_1 y_1 + c_2 y_2$. Exactly n arbitrary constants.

S001·15 · S001·16 · T001 00:24:29 · T001 00:25:14

Graphs and diagrams

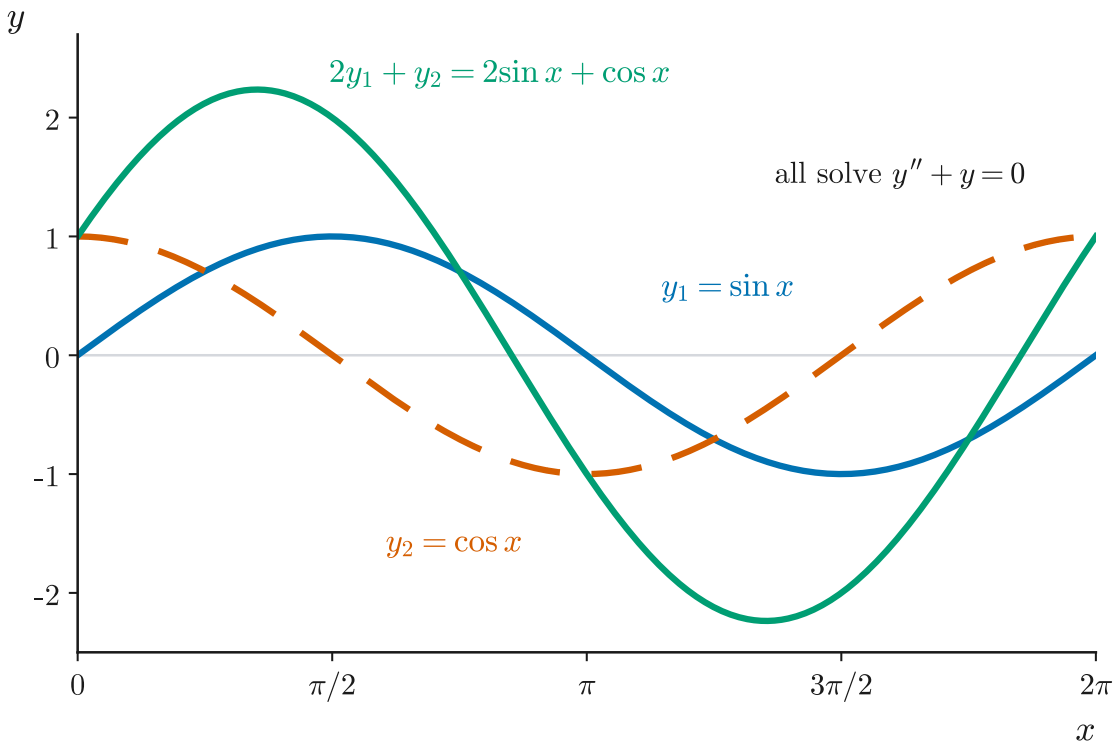


Figure G4.1. Superposition: a combination of solutions is a solution. ESA ★★★

ADDED FOR CLARITY S001·15 · S001·16

How to read it. For $y'' + y = 0$, $y_1 = \sin x$ (blue) and $y_2 = \cos x$ (vermilion dashed) are independent solutions. Their combination $2\sin x + \cos x$ (green) is again a solution; it has a different amplitude and phase, so it is not a multiple of either. Varying c_1, c_2 gives every solution: the CF $c_1 \sin x + c_2 \cos x$ (sympy dsolve).

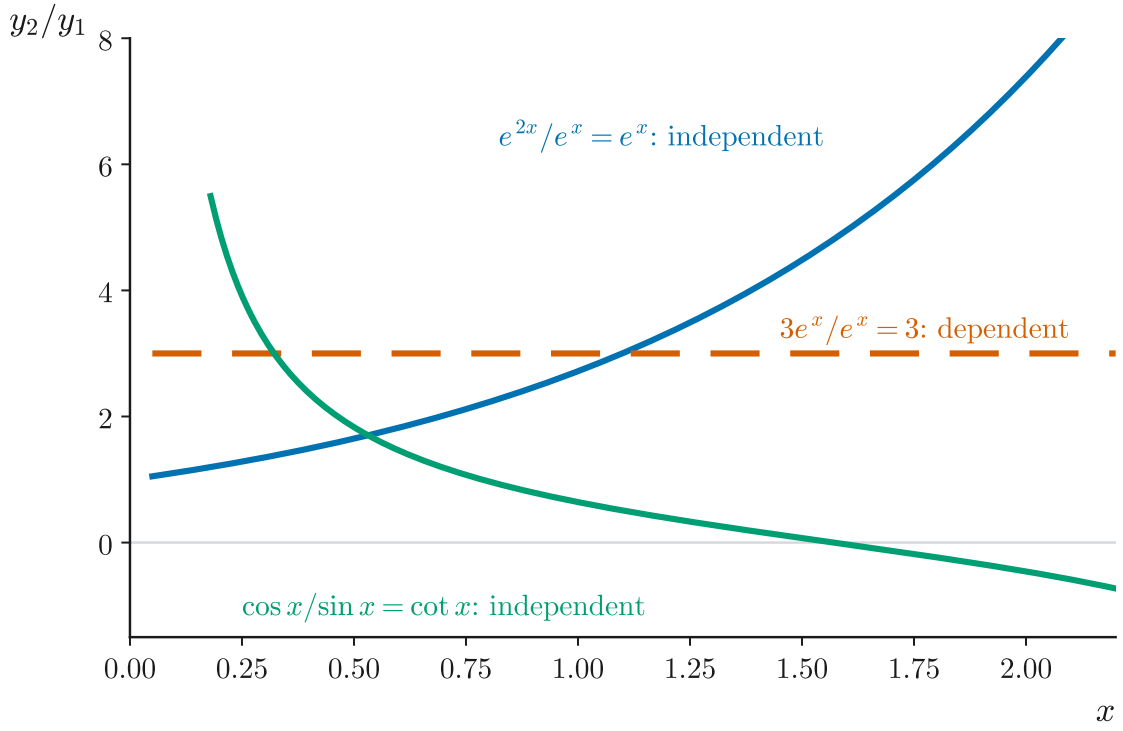
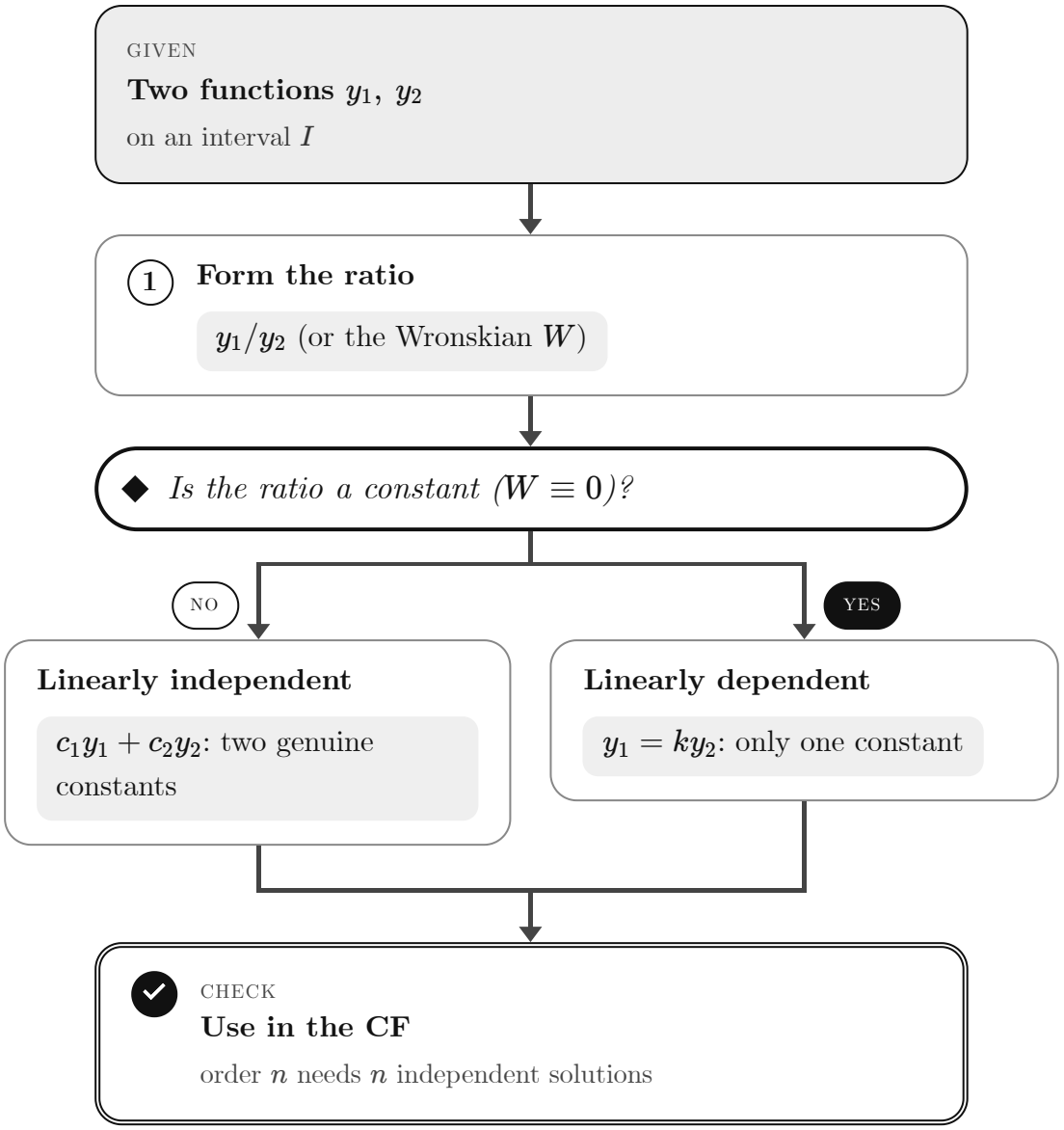


Figure G4.2. The ratio test for two functions. ESA ★★

ADDED FOR CLARITY S001·14 · T001 00:22:00

How to read it. Plot y_2/y_1. For e^x and $3e^x$ the ratio is the constant 3 (vermilion dashed): dependent. For e^x and e^{2x} it is e^x (blue), and for $\sin x$ and $\cos x$ it is $\cot x$ (green): not constant, so independent. A constant ratio is exactly linear dependence.

Flowcharts



Remark. The Wronskian is added for clarity (not in the deck). For a set of n functions use the definition E4.1 or the $n \times n$ Wronskian.

Figure F4.1. Are two functions linearly independent? ESA ★★

ADDED FOR CLARITY S001·13 · S001·14 · T001 00:22:00

Important points and why

1. Two functions are independent when neither is a constant multiple of the other: e^x, e^{2x} and $\sin x, \cos x$ are; $e^x, 3e^x$ are not. ESA ★★ S001·14 · T001 00:22:00
2. Any linear combination of solutions of a homogeneous LDE is again a solution (superposition). With n independent solutions it is the complete solution, the CF. ESA ★★★ S001·15 · S001·16 · T001 00:24:11
3. "Very important concept": linearly independent and dependent functions (T001 00:18:47). Independence guarantees the n constants are genuinely different. ESA ★★ T001 00:18:47 · S001·13

Why must the solutions in the CF be linearly independent?
If $y_2 = ky_1$, then $c_1y_1 + c_2y_2 = (c_1 + kc_2)y_1$ has only one effective constant, so it cannot be the complete solution of a second-order equation, which needs two. Independence makes the n constants genuinely different.
S001·14 · S001·15 · T001 00:22:18

Exam trap. The lecture says $c_1y_1 + c_2y_2$ is a solution "only when y_1 and y_2 are linearly independent" (T001 00:23:40). Wrong: every combination of solutions of a homogeneous LDE is a solution. Independence is what makes it the *complete* solution. ADDED FOR CLARITY T001 00:23:40 · S001·15

Exam trap. Slide 001·14's "$c_1y_1 + c_2y_2 \neq 0$ for non zero constants" must be read as: the combination is not identically zero unless $c_1 = c_2 = 0$. Equivalently y_1/y_2 is not a constant.
ADDED FOR CLARITY S001·14 · T001 00:22:00

Exam trap. Superposition holds for the homogeneous equation only. Two PIs of $f(D)y = X$ do not add to a PI: their sum solves $f(D)y = 2X$. ADDED FOR CLARITY S001·17

Worked examples

W4.1 · Dependent or independent? ESA ★★

GIVEN Test (a) e^x, e^{2x}; (b) $e^x, 3e^x$; (c) $\sin x, \cos x$; (d) $1, x, x^2$.

STEP 1 (a) $e^{2x}/e^x = e^x$, not constant: independent. $W = e^x \cdot 2e^{2x} - e^{2x} \cdot e^x = e^{3x} \neq 0$.

STEP 2 (b) $3e^x/e^x = 3$, constant: $3 \cdot e^x - 1 \cdot 3e^x = 0$ with $c_1 = 3, c_2 = -1$ not zero. Dependent; $W = 0$.

STEP 3 (c) $\tan x$ is not constant; $W = \sin x(-\sin x) - \cos x \cos x = -1 \neq 0$. Independent.

STEP 4 (d) $c_1 + c_2x + c_3x^2 \equiv 0$ forces $c_1 = c_2 = c_3 = 0$ (a nonzero polynomial has finitely many roots); $W = 2$.

ANSWER (a) independent, (b) dependent, (c) independent, (d) independent.

CHECK sympy Wronskians: e^{3x}, 0, -1, 2.

S001·13 · S001·14 · T001 00:22:00

W4.2 · Superposition for $y'' - 3y' + 2y = 0$ ESA ★★★

GIVEN Verify that $y_1 = e^x$ and $y_2 = e^{2x}$ solve $(D^2 - 3D + 2)y = 0$, and that $c_1e^x + c_2e^{2x}$ is the complete solution.

STEP 1 y_1: $e^x - 3e^x + 2e^x = 0$. y_2: $4e^{2x} - 6e^{2x} + 2e^{2x} = 0$.

STEP 2 $(D^2 - 3D + 2)(c_1y_1 + c_2y_2) = c_1 \cdot 0 + c_2 \cdot 0 = 0$ by linearity.

STEP 3 y_1, y_2 independent (W4.1a) and the order is 2, so two constants suffice.

ANSWER $\boxed{y_c = c_1e^x + c_2e^{2x}}$

CHECK sympy dsolve gives the same; substitution gives 0 for all c_1, c_2.

S001·15 · T001 00:23:19

SHORT-NOTE RECAP

$f_1, \ldots, f_n$ are linearly dependent if $c_1f_1 + \cdots + c_nf_n = 0$ with constants not all zero; otherwise independent. Two functions: independent if neither is a constant multiple of the other (e^x, e^{2x} yes; $e^x, 3e^x$ no). If $y_1, \ldots, y_n$ are n independent solutions of the homogeneous LDE, the complete solution $c_1y_1 + \cdots + c_ny_n$ is the complementary function.

5 The general solution: $y = \text{CF} + \text{PI}$

For the non-homogeneous equation $f(D)y = X$, if y_c is the complete solution of $f(D)y = 0$ and y_p is any one solution of $f(D)y = X$, the complete solution is $y = y_c + y_p$. The particular solution y_p is called the *particular integral* (PI). The CF carries all n arbitrary constants; the PI has none. Every Unit 2 ESA question is answered in this form; classes 002–006 give the methods for the CF and the PI.

Key terms and definitions

Particular integral (PI). A particular solution y_p of the non-homogeneous equation $f(D)y = X$: one solution, containing no arbitrary constant. S001·17 · T001 00:27:16 · T001 00:27:52

Complete (general) solution. $y = y_c + y_p$, CF plus PI. S001·17 · T001 00:27:34

> *"The general solution of the non homogeneous linear differential equation has two components: Complementary Function and particular Integral. The complementary function is the complete solution of homogeneous linear differential equation."* S001·12

> *"Consider a non homogeneous linear differential equation $f(D)y = X$. If $y = y_c$ is the complete solution of $f(D)y = 0$ and $y = y_p$ is a particular solution of the differential equation $f(D)y = X$, then the complete solution is $y = y_c + y_p$. The particular solution y_p is called PARTICULAR INTEGRAL."* S001·17

Formulas

E5.1 · Complete solution of $f(D)y = X$ ESA ★★★

$$y = y_c + y_p = \text{CF} + \text{PI}$$

gives	The answer to every non-homogeneous LDE.
applies	y_c: complete solution of $f(D)y = 0$; y_p: any one solution of $f(D)y = X$.
note	All 24 Unit 2 ESA occurrences are answered in this form.

S001·12 · S001·17 · T001 00:27:16 · T001 00:27:34

E5.2 · Why it works (linearity) ESA ★★

$$f(D)(y_c + y_p) = f(D)y_c + f(D)y_p = 0 + X = X$$

gives	$y_c + y_p$ solves $f(D)y = X$ and has all n constants.
applies	$f(D)$ linear.
note	Also: the PI of $X_1 + X_2$ is the sum of the PIs of X_1 and X_2 (W5.3).

added for clarity S001·17

Graphs and diagrams

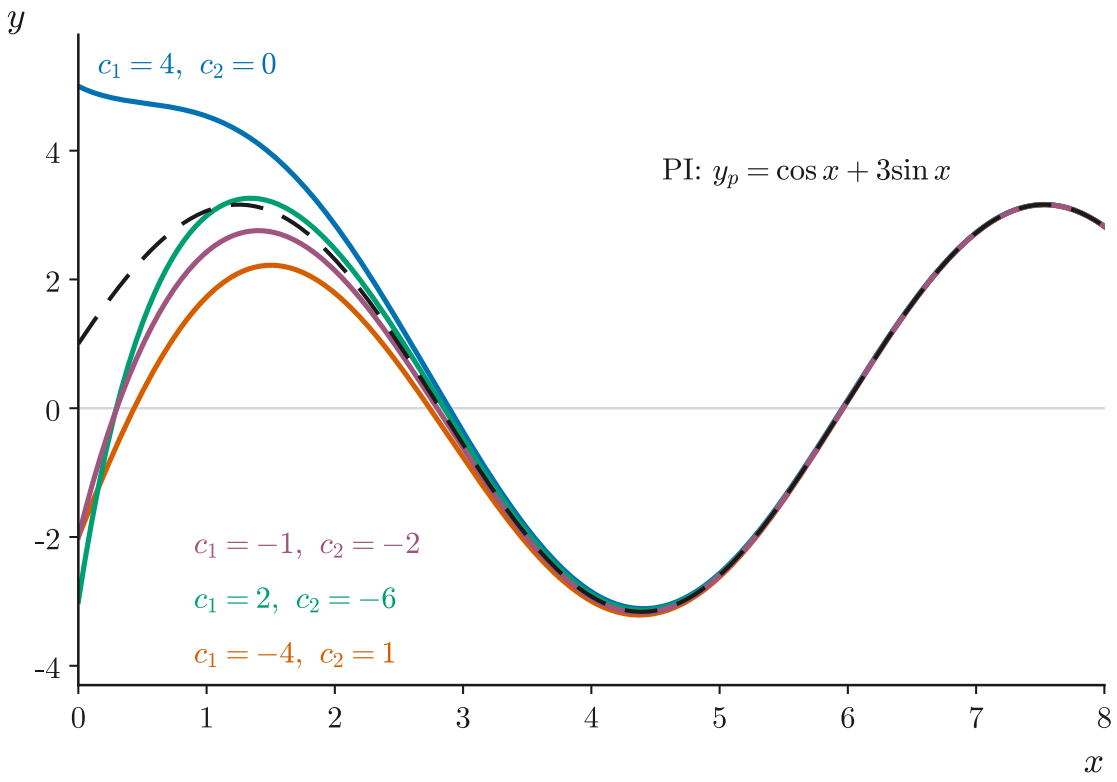
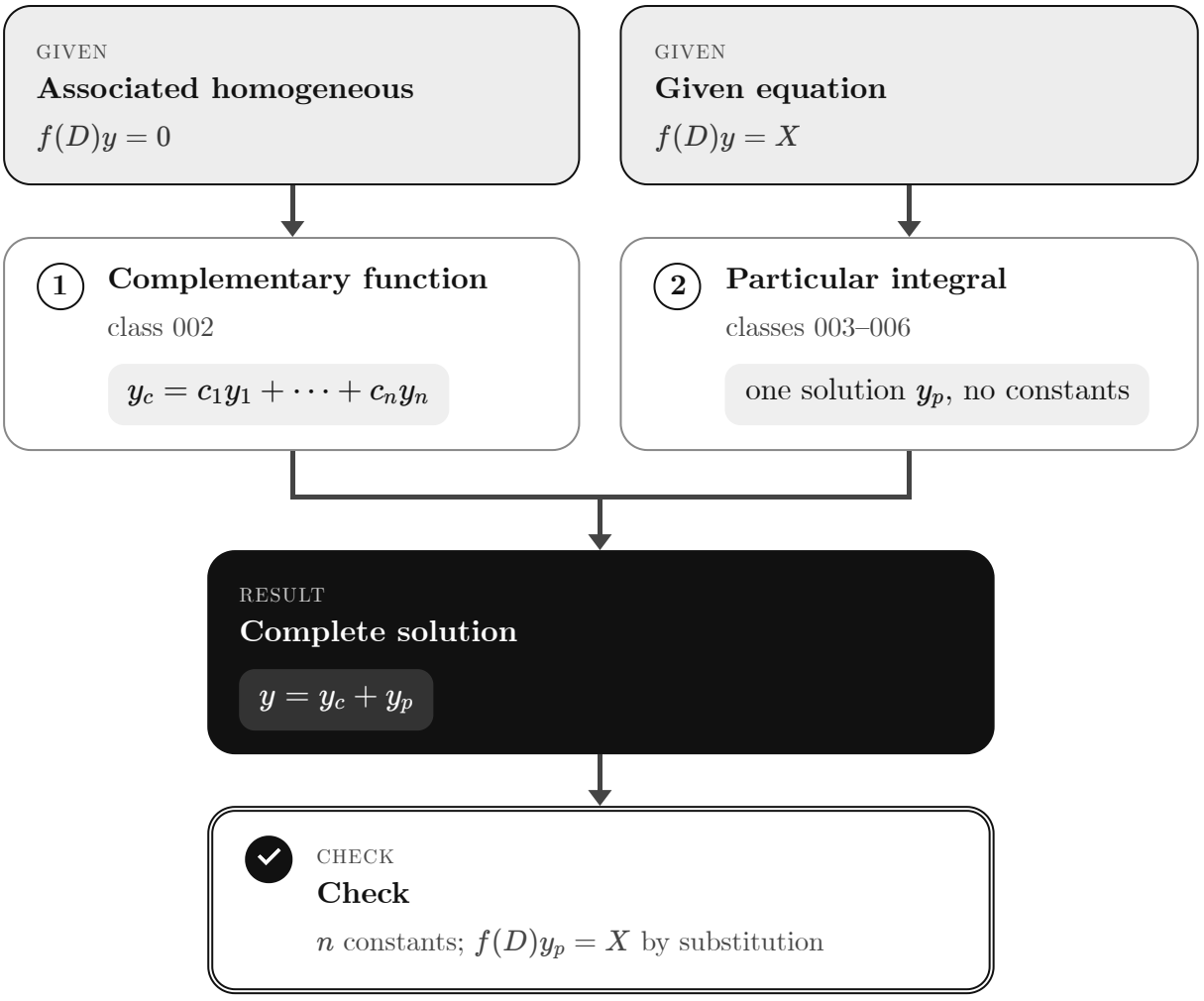


Figure G5.1. Every solution is CF + PI: one PI, many CFs. ESA ★★★

ADDED FOR CLARITY S001·17 · S001·12

How to read it. For $y'' + 3y' + 2y = 10\cos x$ the PI is $y_p = \cos x + 3\sin x$ (black dashed) and the CF is $c_1 e^{-x} + c_2 e^{-2x}$. Four choices of c_1, c_2 (coloured curves) start at different values but all approach y_p, because here the CF decays. The constants come from the CF only; the PI is the same for all. Every curve was checked with sympy dsolve.

Flowcharts



Remark. Initial conditions, if given, are applied to the complete solution $y_c + y_p$, never to the CF alone.

Figure F5.1. The plan of every Unit 2 answer ESA ★★★

S001·12 · S001·17 · T001 00:18:29 · ESA Dec-23 3a

Complementary function against particular integral

	Complementary function y_c	**Particular integral y_p**
Solves	$f(D)y = 0$ (associated homogeneous)	$f(D)y = X$ (given equation)
Arbitrary constants	n (order of the DE)	none
Form	$c_1y_1 + \cdots + c_ny_n$, y_i independent	one function of x
Unique?	the complete family	no: any two differ by part of the CF
Depends on	$f(D)$ only	$f(D)$ and X
Found in	class 002 (auxiliary equation)	classes 003–006, 011 (variation of parameters)

S001·12 · S001·16 · S001·17

Every Unit 2 ESA question is CF + PI

Kind of question	**Distinct (occurrences)**	**Coefficients**	**Answer form**
PI by operator rules	7 (9)	constant, $f(D)y = X$	CF + PI
Legendre / Cauchy	6 (8)	variable; reduced to constant	CF + PI in z, then back to x
Variation of parameters	4 (5)	constant	CF + PI (PI from W)
LCR circuits	2 (2)	constant, in t	charge = CF (transient) + PI (steady state)
Total	19 (24)		all CF + PI

ESA Dec-23 3a · ESA Jul-23 3c · ESA Jan-24 s3 3b · ESA Jan-24 s3 3d · ESA Dec-23 3c

Important points and why

1. Every Unit 2 ESA question (19 distinct, 24 occurrences in 8 papers) is a non-homogeneous linear DE whose answer is $y = \mathbf{CF + PI}$. Write both parts and add them; an answer with only the CF or only the PI is incomplete. ESA ★★★ S001·17 · ESA Dec-23 3a · ESA Jul-23 3b · ESA Jan-24 s2 2d
2. The CF holds all n arbitrary constants of an n-th order equation; the PI has none. Count the constants as a check. ESA ★★★ S001·16 · S001·17 · T001 00:25:38
3. The CF is the complete solution of the associated homogeneous equation $f(D)y = 0$ (put $X = 0$); find it first. ESA ★★★ S001·12 · T001 00:17:58 · T001 00:26:57

Why is $y_c + y_p$ the complete solution of $f(D)y = X$?
$f(D)$ is linear: $f(D)(y_c + y_p) = 0 + X$. Conversely, if y is any solution, $f(D)(y - y_p) = 0$, so $y - y_p$ is a solution of the homogeneous equation and is contained in y_c. Hence every solution has the form $y_c + y_p$.

ADDED FOR CLARITY S001·17

Why is the PI called "particular"?
It is one particular solution, with no arbitrary constants. Any other choice differs from it by a solution of $f(D)y = 0$, which the CF absorbs: $\frac{1}{2}e^{3x}$ and $\frac{1}{2}e^{3x} + e^{x}$ are both PIs of $y'' - 3y' + 2y = e^{3x}$ (W5.1).

ADDED FOR CLARITY S001·17 · T001 00:27:34

Exam trap. Put the constants in the CF only; the PI has none. Do not fit initial conditions to the CF alone: apply them to $y = y_c + y_p$. ADDED FOR CLARITY S001·17

Worked examples

W5.1 · Check a CF + PI solution — ESA ★★★

GIVEN Verify that $y = c_1e^{x} + c_2e^{2x} + \frac{1}{2}e^{3x}$ is the complete solution of $(D^2 - 3D + 2)y = e^{3x}$.

STEP 1 CF: $c_1e^{x} + c_2e^{2x}$ solves $(D^2 - 3D + 2)y = 0$ (W4.2).

STEP 2 PI: $y_p = \frac{1}{2}e^{3x}$: $\frac{9}{2}e^{3x} - \frac{9}{2}e^{3x} + e^{3x} = e^{3x}$ ✓.

STEP 3 Sum: $f(D)(y_c + y_p) = 0 + e^{3x}$; two constants for order 2.

ANSWER $y = c_1e^{x} + c_2e^{2x} + \frac{1}{2}e^{3x}$ (CF + PI)

CHECK sympy dsolve agrees. $\frac{1}{2}e^{3x} + e^{x}$ is also a PI; the extra e^{x} is absorbed in c_1.

ADDED FOR CLARITY S001·17

W5.2 · Anatomy of an ESA answer (ESA Dec-23 3a) ESA ★★★

GIVEN Solve $(D-2)^2y = 8(e^{2x} + \sin 2x)$ (asked 3 times). Identify the CF, the PI and the complete solution.

STEP 1 Associated homogeneous: $(D-2)^2y = 0$; the root 2 repeats, so $y_c = (c_1 + c_2x)e^{2x}$ (class 002).

STEP 2 PI (classes 003–004): $y_p = 4x^2e^{2x} + \cos 2x$. Check: $(D-2)^2(4x^2e^{2x}) = 8e^{2x}$ and $(D^2 - 4D + 4)\cos 2x = 8\sin 2x$.

STEP 3 Add: $y = y_c + y_p$.

ANSWER $\boxed{y = (c_1 + c_2x)e^{2x} + 4x^2e^{2x} + \cos 2x}$

CHECK sympy dsolve and checkodesol agree. The PI methods are in the class 004 pack; here only the structure matters.

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a

W5.3 · PI of a sum is the sum of PIs (ESA Jul-23 3b) ESA ★★★

GIVEN For $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$, verify $y_{p1} = \frac{1}{2}e^{4x}$, $y_{p2} = \frac{1}{10}(\sin x + \cos x)$, $y_{p3} = \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$ for the three terms, and write the complete solution.

STEP 1 $f(D)y_{p1} = e^{4x}$, $f(D)y_{p2} = \sin x$, $f(D)y_{p3} = x^2$ (substitute each; sympy).

STEP 2 Linearity: $f(D)(y_{p1} + y_{p2} + y_{p3}) = e^{4x} + \sin x + x^2$, so the PI is the sum.

STEP 3 CF: $f(D) = (D-2)(D-3)$, $y_c = c_1e^{2x} + c_2e^{3x}$.

ANSWER $\boxed{y = c_1e^{2x} + c_2e^{3x} + \frac{1}{2}e^{4x} + \frac{1}{10}(\sin x + \cos x) + \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}}$

CHECK sympy dsolve and checkodesol agree. How each PI is found: class 005 pack.

ESA Jul-23 3b

Past-paper questions

ESA Dec 2023 · 7 marks · UE22MA141A Q3.a (also May 2024 Q3.b, June 2024 Q2.a) · solved in W5.2

Solve $(D-2)^2y = 8(e^{2x} + \sin 2x)$. (Here: the CF + PI structure.)

ESA Jul 2023 · 7 marks · UE22MA141A Q3.b · solved in W5.3

Solve $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$. (Here: the PI of a sum.)

SHORT-NOTE RECAP

For $f(D)y = X$: $y = y_c + y_p$, where y_c (CF) is the complete solution of $f(D)y = 0$ and y_p (PI) is one particular solution of $f(D)y = X$. CF: n constants; PI: none. Linearity gives $f(D)(y_c + y_p) = X$, and the PI of a sum of terms is the sum of their PIs. Every Unit 2 ESA answer has this form.

A Symbol glossary

Symbol	Name	Meaning	Appears in
x	Independent variable	The variable we differentiate with respect to	all
y	Dependent variable	The unknown function $y(x)$	all
X	Right-hand side	A function of x alone; $X = 0$ homogeneous. Not the variable x	E1.2, E1.3, E3.2
n	Order	Highest derivative; also the number of constants in the CF	E1.3, E4.4
P, Q	First-order coefficients	Functions of x in $y' + Py = Q$	E1.1
$P_1, \ldots, P_n$	Coefficients (slide 001·7)	Functions of x; the lecture adds $P_0 \neq 0$	E1.2
$a_1, \ldots, a_n$	Coefficients (slides 001·8 on)	Functions of x or constants	E1.3, E3.2
k_1, k_2	Constant coefficients	In $y'' + k_1 y' + k_2 y = 0$ (slide 001·15)	S001·15
D	Differential operator	$\frac{d}{dx}$; $D^n = \frac{d^n}{dx^n}$	E3.1
$f(D)$	Linear differential operator	$D^n + a_1 D^{n-1} + \cdots + a_n$	E3.2, E5.1
$f_1, \ldots, f_n$	Functions	Tested for linear dependence	E4.1
$c_1, \ldots, c_n$	Constants	In a linear combination; the arbitrary constants of the CF	E4.1, E4.4
$y_1, \ldots, y_n$	Independent solutions	Of the homogeneous LDE	E4.4
W	Wronskian	$y_1 y_2' - y_2 y_1'$ (added for clarity)	E4.3
y_c	Complementary function	Complete solution of $f(D)y = 0$	E4.4, E5.1
y_p	Particular integral	One solution of $f(D)y = X$	E5.1
m	Root of the auxiliary equation	$f(m) = 0$ (preview of class 002)	E3.4

B Sources, coverage and gaps

Source	Path	Use
Deck 001 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.*	18 slides; content on 5–17 (every equation read from the rendered images; the text files have none)
Transcript 001 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.*	28 min; slides 3–17; no low-confidence segment
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	No question on this class alone; the CF + PI structure of every Unit 2 question, via the shared ESA map
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Checked: no Unit 2 problem belongs to this class (problems 1–9 are class 002)
Video 001 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.mp4	Not opened: every slide is readable and the transcript has no low-confidence segment

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question targets this class directly (definitions, classification, linear independence and the Wronskian were never asked on their own: 0 of 8 papers). But every Unit 2 ESA question (19 distinct questions, 24 occurrences) is a non-homogeneous linear DE whose answer is $y = \text{CF} + \text{PI}$, and every constant-coefficient one is written in D-notation $f(D)y = X$. So the structure $y = \text{CF} + \text{PI}$ and the $f(D)$ notation are ★★★; recognising constant against variable coefficients (8 Legendre/Cauchy occurrences need it) and homogeneous against non-homogeneous ★★; definitions and the linear-dependence test ★ (never asked; the Wronskian is used in class 011, variation of parameters).

Known gaps, conflicts and corrections

1. Slide 001·14 says y_1, y_2 are linearly independent "when $c_1y_1 + c_2y_2 \neq 0$ for non zero constants c_1 and c_2". Precisely: $c_1y_1 + c_2y_2 = 0$ for all x in I only when $c_1 = c_2 = 0$; equivalently, neither function is a constant multiple of the other (as the lecture says at T001 00:22:00). Recorded as an exam trap.
2. The lecture says "$c_1y_1 + c_2y_2$ is a solution only when y_1 and y_2 are linearly independent" (T001 00:23:40). That is wrong: any linear combination of solutions of a homogeneous linear DE is a solution. Independence is what makes it the complete (general) solution. Slide 001·15 is correct. Recorded as an exam trap.
3. The lecture's "P_0 equal to 0" for the second-order form (T001 00:07:34) must be $P_0 \neq 0$, as it says for order n (T001 00:08:55); otherwise the equation is not of order 2.

4. The lecture's second definition of a linear DE (T001 00:06:27) requires the coefficients of y and its derivatives to be functions of x or constants; the coefficient must not involve y. Stated precisely in the definitions.
5. The lecture says $f(D)$ "obeys the laws of algebra" and its factors "can be written in any order" (T001 00:16:49). True for constant coefficients only; with variable coefficients $D(xD) \neq (xD)D$. Added as an exam trap, flagged added for clarity.
6. The Wronskian, the nonzero-determinant test for independence, is not in the deck or lecture; it is added for clarity (formula E4.3) because class 011 (variation of parameters) uses it.
7. The auxiliary equation (E3.4) is only previewed here; it is not in this deck. Every CF case is in the class 002 pack.
8. The deck has no worked problems and the CW sheet has none for this class; the worked examples use the deck's equations and two ESA questions (Dec-23 3a, Jul-23 3b), solved here only to show the CF + PI structure (sympy dsolve and checkodesol).
9. ESA: no question on this class alone. The shared ESA map gives 19 distinct Unit 2 questions and 24 occurrences; all are non-homogeneous and answered as CF + PI.
10. Bucket session numbers are not contiguous (decks say SESSION 1–11; bucket 009 = session 7, 011 VoP = 8, 011 Legendre = 9, 014 = 10, 015 = 11; 007, 008, 010, 012, 013 do not exist).
11. Transcript recognition errors were corrected silently: "dyed by dx" → $\frac{dy}{dx}$, "d by ts" → $\frac{d}{dx}$, "Albert 2" / "our del 3" / "R n" → order 2 / order 3 / order n, "n third derivative" → n-th derivative, "capital S" → X, "particular invariant" / "particular integration" → particular integral, "variance" → variable, "a 10 function" → a nonzero function, "minus p" → -5, "c1 by 1"/"C1 by 1" → $c_1 y_1$, "f of d of i" → $f(D)y$, "E naught even into A N" → $a_0, a_1, \ldots, a_n$, "k2 div" → $k_2 y$.

Coverage

Source	Citations	Slides cited (slide × citations)
S001	98	3×1, 5×4, 6×3, 7×7, 8×8, 9×6, 10×7, 11×9, 12×8, 13×6, 14×9, 15×7, 16×8, 17×15
ESA	17	
T001	80	

Slides 001·1, 001·2 and 001·18 are title, session and closing slides; 001·3 lists the unit content and 001·4 the class content. The lecture (28 min) covers every content slide 5–17 in order. The deck has no worked problems and no class-work problem is assigned to this class, so the worked examples use the deck's own equations plus two ESA questions read for their structure only. Bucket session numbers are not contiguous: the decks say SESSION 1–11, but the bucket numbers them 001–006, 009, 011 (two decks), 014 and 015; 007, 008, 010, 012 and 013 do not exist. This deck is SESSION 1 in both numberings.